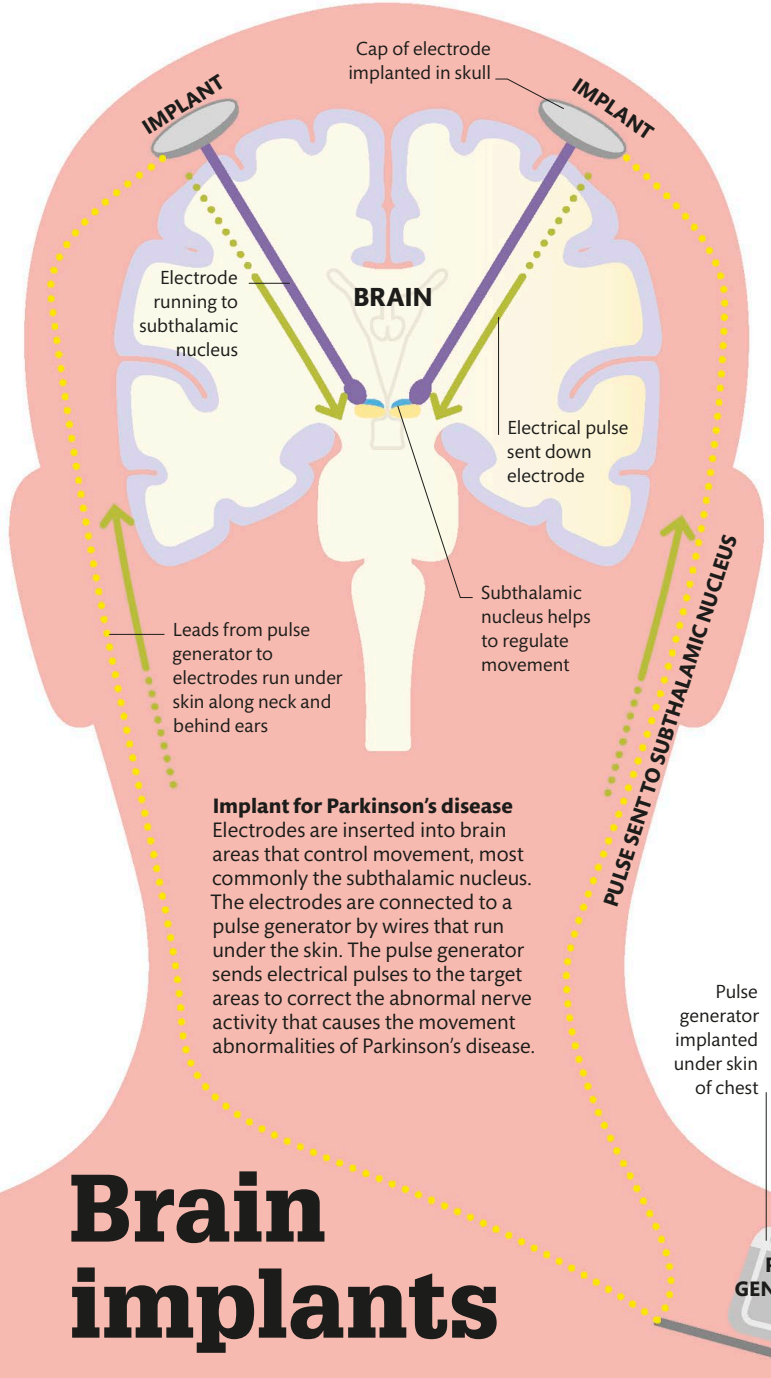
Brain implants

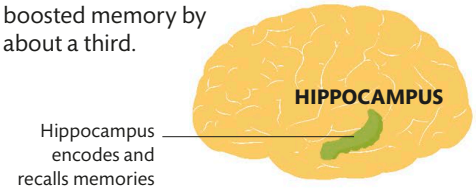
A brain implant is an artificial device embedded in the brain that works with one or more other devices to improve or restore brain function in people disabled by injury or illness. A sensory implant interfaces with the brain via the nervous system and may help to restore hearing or vision. The technology of implants is still in its early stages.



Implant for Parkinson's disease
Electrodes are inserted into brain areas that control movement, most commonly the subthalamic nucleus. The electrodes are connected to a pulse generator by wires that run under the skin. The pulse generator sends electrical pulses to the target areas to correct the abnormal nerve activity that causes the movement abnormalities of Parkinson's disease.

MEMORY IMPLANTS

Scientists are developing brain implants to improve memory. In one study, people with epilepsy who already had brain implants had electrodes inserted into an area of the brain called the hippocampus. Their brain signals were recorded while they completed memory tests. Later, the same brain signals were used to stimulate their brains while they carried out similar tests. The stimulation boosted memory by about a third.



Deep brain stimulation

Stimulation of specific groups of nerve cells deep in the brain—known as deep brain stimulation, or DBS—may be used to restore normal brain activity in people with Parkinson's disease, certain other movement disorders, or epilepsy. Electrodes are implanted in the brain, and a device called a pulse generator, implanted in the chest or stomach, emits electrical pulses to regulate the brain's activity. The device may operate continuously or only when electrodes detect abnormal nerve signals (such as if an epileptic seizure starts). After the system has been fitted, a specialist programs the pulse generator so that it produces pulses only when necessary.

WHAT ARE BRAIN ELECTRODES MADE OF?

Electrodes implanted in the brain are made of substances such as gold or platinum-iridium, which conduct electrical impulses well and do not harm brain tissue.